

Vanshika Sharma

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Profile

AIML undergraduate at IGDTUW (CGPA: 8.69) building full-stack applications using Django, FastAPI, React, PostgreSQL. Built production-grade systems including privacy-preserving cognitive health platform and ML based perceptual search engine. Experienced in backend engineering, REST APIs, testing and database optimization.

Education

Indira Gandhi Delhi Technical University for Women

Aug 2023 – May 2027

B.Tech in Artificial Intelligence and Machine Learning

Cumulative CGPA: 8.69

Experience

ML Research Intern

June 2025 – Aug 2025

Department of Artificial Intelligence and Data Science

IGDTUW, New Delhi

- Developed a phishing detection system with 92% accuracy using feature engineering and cross-validation.
- Benchmarked Random Forest, SVM, and PCA pipelines, improving model accuracy by 15% across evaluation sets.

Software Developer Intern

June 2024 – Aug 2024

Clovio Pvt Ltd

Noida, Uttar Pradesh

- Built role-based access control for 3 user types using Django AbstractUser, custom permissions, and protected views.
- Validated system reliability with 51 automated tests covering models, views, form validation, and access control.
- Optimized database performance removing N+1 queries & authentication with CSRF-protected POST-only flows.

Projects

Kogni | *Chrome Extension, FastAPI, SQLite, scikit-learn, SHAP, Next.js, WebSockets*

May 2026 – June 2026

- Built privacy-preserving cognitive health platform with a Chrome Extension (Manifest V3) that captures keystroke timing and scroll behavior as anonymized feature vectors; raw browsing history never leaves the browser.
- Trained RF model on Dartmouth CES (210K+ datapoints) for cognitive fatigue scoring with SHAP explainability, and implemented a bidirectional LSTM in NumPy achieving 90.8% accuracy on cognitive trajectory prediction.
- Developed a recovery engine comparing post-exercise IKI against 7-day personal baselines to detect cognitive restoration (Inspired by the paper arXiv:2509.23158), and deployed full-stack infrastructure.

Persens | *PyTorch, CLIP, MLP, FastAPI, Qdrant, React, TypeScript, Docker*

March 2026 – May 2026

- Extended Filip et al.'s (arXiv:2410.13615) material fingerprinting system with attribute-directed perceptual search, achieving Recall@5 = 3.2/5 on 347 materials validated by 110,000+ human ratings; retrieval fidelity MAE = 0.12.
- Predicted material appearance using neural networks; reduced error by 31% over baseline across 1,000 test cases.
- Implemented per-attribute Grad-CAM across 16 perceptual dimensions and material authentication achieving ROC-AUC = 0.89; deployed as FastAPI + Qdrant API with React frontend at persens.vercel.app.

Exit Management System | *Python, Django, Django ORM, SQLite, HTML/CSS*

June 2024 – Sep 2024

- Role-based offboarding platform for HR, HOD & Employee with a 5-state cycle (Initiated → In Progress → Cleared) that auto-progresses based on task completion and interview submission across all departments.
- Departmental clearance tasks auto-assigned on exit initiation; each HOD manages department's tasks across all exits.
- Append-only audit log records every action with actor, timestamp, and old/new values, immutable via save() override

Technical Skills

Languages: Python, C/C++, Java, SQL, JavaScript, HTML, CSS, TypeScript

Frameworks: React, Node.js, Next.js, Flask, FastAPI, Django, JUnit, WebSockets

Tools: Git, Docker, SQLite, Chrome Extension (Manifest V3), Firebase, VS Code, PostgreSQL

Libraries: Pandas, NumPy, Matplotlib, PyTorch, CLIP, SHAP, scikit-learn

Achievements

Awarded **Nutanix WIT Scholarship 2025** (1 of only 4 women in India).

Selected as **Millennium Fellow 2025**, impacting 500+ beneficiaries and aligned with the UN SDGs.

Explored Web3, blockchain, and DeFi as **SheFi Scholar 2024**, through a global fellowship cohort.

Leadership Experience

Vice President | *180 Degrees Consulting, IGDTUW*

Feb 2026 – Present

- Won **Mappls Business Hackathon, BITS Pilani** and **Top 5 in Global Case Competition**, IIT Kanpur.
- Led a 45+ member team delivering solutions for startups/organizations like Haqdarshak, Sonetz, KBG Induction.

UI/UX Circle Mentor | *Google Developer Groups, IGDTUW*

Oct 2024 – Oct 2025

- Mentored 80+ students in UI/UX design, fostering skill development and professional growth.
- Conducted 15+ hands-on workshops, boosting project completion and engagement rates by 30%.

Project Leader | *Enactus, IGDTUW*

Sep 2024 – Sep 2025

- Launched 2 MVPs using 120+ repurposed lithium-ion cells for EV accessories and backup power.
- Gained funding/ recognition in 5+ National B-Plan competitions at IIT Roorkee, DTU, NSUT, IMT Nagpur.